

## MCU\_A

## U1A

|         |           |      |     |                                                               |
|---------|-----------|------|-----|---------------------------------------------------------------|
| GBC_KEY | WK_UP     | PA0  | 34  | PA0-WKUP/USART2_CTS/ADC123_IN0/TIM5_CH1/TIM2_CH1_ETR/TIM8_ETR |
|         | STM_ADC   | PA1  | 35  | PA1/USART2_RTS/ADC123_IN1/TIM5_CH2/TIM2_CH2                   |
|         | USART2_TX | PA2  | 36  | PA2/USART2_TX/ADC123_IN2/TIM5_CH3/TIM2_CH3                    |
|         | USART2_RX | PA3  | 37  | PA3/USART2_RX/ADC123_IN3/TIM5_CH4/TIM2_CH4                    |
|         | STM_DAC   | PA4  | 40  | PA4/SPI1_NSS/DAC_OUT1/USART2_CK/ADC12_IN4                     |
|         |           | PA5  | 41  | PA5/SPI1_SCK/DAC_OUT2/ADC12_IN5                               |
|         |           | PA6  | 42  | PA6/SPI1_MISO/TIM8_BKIN/ADC12_IN6/TIM3_CH1                    |
|         |           | PA7  | 43  | PA7/SPI1_MOSI/TIM8_CH1N/ADC12_IN7/TIM3_CH2                    |
|         | OV_VSYNC  | PA8  | 100 | PA8/USART1_CK/TIM1_CH1/MCO                                    |
|         | USART1_TX | PA9  | 101 | PA9/USART1_TX/TIM1_CH2                                        |
|         | USART1_RX | PA10 | 102 | PA10/USART1_RX/TIM1_CH3                                       |
|         | USB_D-    | PA11 | 103 | PA11/USART1_CTS/CAN_RX/TIM1_CH4/USBDM                         |
|         | USB_D+    | PA12 | 104 | PA12/USART1_RTS/CAN_TX/TIM1_ETR/USBBDP                        |
|         | JTMS      | PA13 | 105 | PA13/JTMS_SWDIO                                               |
| GBC_LED | JTCK      | PA14 | 109 | PA14/JTCK_SWCLK                                               |
|         | JTDI      | PA15 | 110 | PA15/JTDI/SPI3_NSS/I2S3_WS                                    |

|         |           |      |     |                                                     |
|---------|-----------|------|-----|-----------------------------------------------------|
|         | LCD_BL    | PB0  | 46  | PB0/ADC12_IN8/TIM3_CH3/TIM8_CH2N                    |
| T_MISO  | T_SCK     | PB1  | 47  | PB1/ADC12_IN9/TIM3_CH4/TIM8_CH3N                    |
|         | BOOT1     | PB2  | 48  | PB2/BOOT1                                           |
| OV_WEN  | JTDO      | PB3  | 133 | PB3/JTDO/TRACESWO/SPI3_SCK/I2S3_CK                  |
| OV_RCLK | JTRST     | PB4  | 134 | PB4/JNTRST/SPI3_MISO                                |
|         | LED0      | PB5  | 135 | PB5/I2C1_SMBAL/SPI3_MOSI/I2S3_SD                    |
|         | IIC_SCL   | PB6  | 136 | PB6/I2C1_SCL/TIM4_CH1                               |
|         | IIC_SDA   | PB7  | 137 | PB7/I2C1_SDA/FSMC_NADV/TIM4_CH2                     |
|         | BEEP      | PB8  | 139 | PB8/TIM4_CH3/SDIO_D4                                |
|         | REMOTE_IN | PB9  | 140 | PB9/TIM4_CH4/SDIO_D5                                |
|         | GBC_RX    | PB10 | 69  | PB10/I2C2_SCL/USART3_TX                             |
|         | GBC_TX    | PB11 | 70  | PB11/I2C2_SDA/USART3_RX                             |
|         | F_CS      | PB12 | 73  | PB12/SPI2_NSS/I2S2_WS/I2C2_SMBAL/USART3_CK/TIM1BKIN |
|         | SPI2_SCK  | PB13 | 74  | PB13/SPI2_SCK/I2S2_CK/USART3_CTS/TIM1_CH1N          |
|         | SPI2_MISO | PB14 | 75  | PB14/SPI2_MISO/USART3_RTS/TIM1_CH2N                 |
|         | SPI2_MOSI | PB15 | 76  | PB15/SPI2_MOSI/I2S2_SD/TIM1_CH3N                    |

|          |      |     |                               |
|----------|------|-----|-------------------------------|
| OV_D0    | PC0  | 26  | PC0/ADC123_IN10               |
| OV_D1    | PC1  | 27  | PC1/ADC123_IN11               |
| OV_D2    | PC2  | 28  | PC2/ADC123_IN12               |
| OV_D3    | PC3  | 29  | PC3/ADC123_IN13               |
| OV_D4    | PC4  | 44  | PC4/ADC12_IN14                |
| OV_D5    | PC5  | 45  | PC5/ADC12_IN15                |
| OV_D6    | PC6  | 96  | PC6/I2S2_MCK/TIM8_CH1/SDIO_D6 |
| OV_D7    | PC7  | 97  | PC7/I2S3_MCK/TIM8_CH2/SDIO_D7 |
| SDIO_D0  | PC8  | 98  | PC8/TIM8_CH3/SDIO_D0          |
| SDIO_D1  | PC9  | 99  | PC9/TIM8_CH4/SDIO_D1          |
| SDIO_D2  | PC10 | 111 | PC10/UART4_TX/SDIO_D2         |
| SDIO_D3  | PC11 | 112 | PC11/UART4_RX/SDIO_D3         |
| SDIO_SCK | PC12 | 113 | PC12/UART5_TX/SDIO_CK         |
|          | PC13 | 7   | PC13-TAMPER-RTC               |
|          |      | 8   | PC14-OSC32_IN                 |
|          |      | 9   | PC15-OSC32_OUT                |

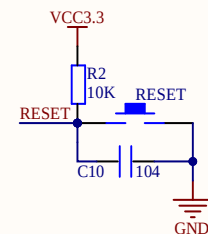
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|----------|------|-----|--------------------------------|
| FSMC_D2  | PD0  | 114 | PD0/FSMC_D2                    |
| FSMC_D3  | PD1  | 115 | PD1/FSMC_D3                    |
| SDIO_CMD | PD2  | 116 | PD2/TIM3_ETR/UART5_RX/SDIO_CMD |
| OV_SCL   | PD3  | 117 | PD3/FSMC_CLK                   |
| FSMC_NOE | PD4  | 118 | PD4/FSMC_NOE                   |
| FSMC_NWE | PD5  | 119 | PD5/FSMC_NWE                   |
| OV_WRST  | PD6  | 122 | PD6/FSMC_NWAIT                 |
| RS485_RE | PD7  | 123 | PD7/FSMC_NE1/FSMC_NCE2         |
| FSMC_D13 | PD8  | 77  | PD8/FSMC_D13                   |
| FSMC_D14 | PD9  | 78  | PD9/FSMC_D14                   |
| FSMC_D15 | PD10 | 79  | PD10/FSMC_D15                  |
|          | PD11 | 80  | PD11/FSMC_A16                  |
|          | PD12 | 81  | PD12/FSMC_A17                  |
|          | PD13 | 82  | PD13/FSMC_A18                  |
| FSMC_D0  | PD14 | 85  | PD14/FSMC_D0                   |
| FSMC_D1  | PD15 | 86  | PD15/FSMC_D1                   |

STM32F103ZET6\_AB

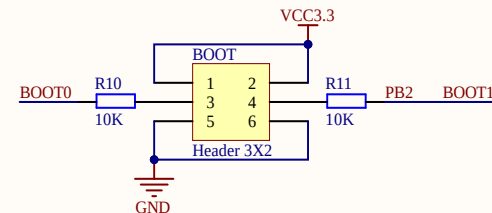
## LCD

|          |        |        |        |         |          |         |          |  |  |
|----------|--------|--------|--------|---------|----------|---------|----------|--|--|
|          |        |        |        | TFT_LCD |          |         |          |  |  |
| FSMC_NE4 | 1      |        |        | 2       | FSMC_A10 |         |          |  |  |
| FSMC_NWE | 3      | CS     | RS     | 4       | FSMC_NOE |         |          |  |  |
| RESET    | 5      | WR     | RD     | 6       | PD14     | FSMC_D0 |          |  |  |
| FSMC_D1  | PD15   | 7      | RST    | DB0     | 8        | PD0     | FSMC_D2  |  |  |
| FSMC_D3  | PD1    | 9      | DB1    | DB2     | 10       | PE7     | FSMC_D4  |  |  |
| FSMC_D5  | PE8    | 11     | DB3    | DB4     | 12       | PE9     | FSMC_D6  |  |  |
| FSMC_D7  | PE10   | 13     | DB5    | DB6     | 14       | PE11    | FSMC_D8  |  |  |
| FSMC_D9  | PE12   | 15     | DB7    | DB8     | 16       | PE13    | FSMC_D10 |  |  |
| FSMC_D11 | PE14   | 17     | DB9    | DB10    | 18       | PE15    | FSMC_D12 |  |  |
| FSMC_D13 | PD8    | 19     | DB11   | DB12    | 20       | PD9     | FSMC_D14 |  |  |
| FSMC_D15 | PD10   | 21     | DB13   | DB14    | 22       |         |          |  |  |
| LCD_BL   | 23     | DB15   | GND    | 24      | VCC3.3   | C1      | 104      |  |  |
| VCC3.3   | 25     | BL     | VDD3.3 | 26      | VCC3.3   | C1      | 104      |  |  |
|          | 27     | VDD3.3 | GND    | 28      | VCC5     | C2      | 104      |  |  |
| PB2      | GND    | 29     | GND    | 30      | PF9      | T_MOSI  |          |  |  |
| PF10     | T_MISO | 31     | MISO   | 32      | PF9      | T_MOSI  |          |  |  |
| PF11     | T_PEN  | 33     | T_PEN  | 34      | PB1      | T_SCK   |          |  |  |
|          | T_CS   | 33     | T_CS   | CLK     |          |         |          |  |  |
|          |        |        |        | TFT_LCD |          |         |          |  |  |

## RESET



## BOOT



Title:

ALIENTEK Elite STM32F103 Board

Author:

ATOM

Date:

2025/11/3

Revision:

\*

Size:

A4

File:

CORE\_ABCD.SchDoc

Version:

V2.6

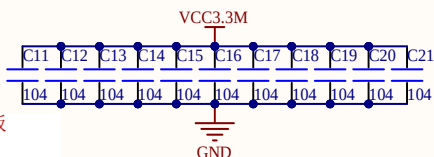
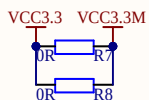
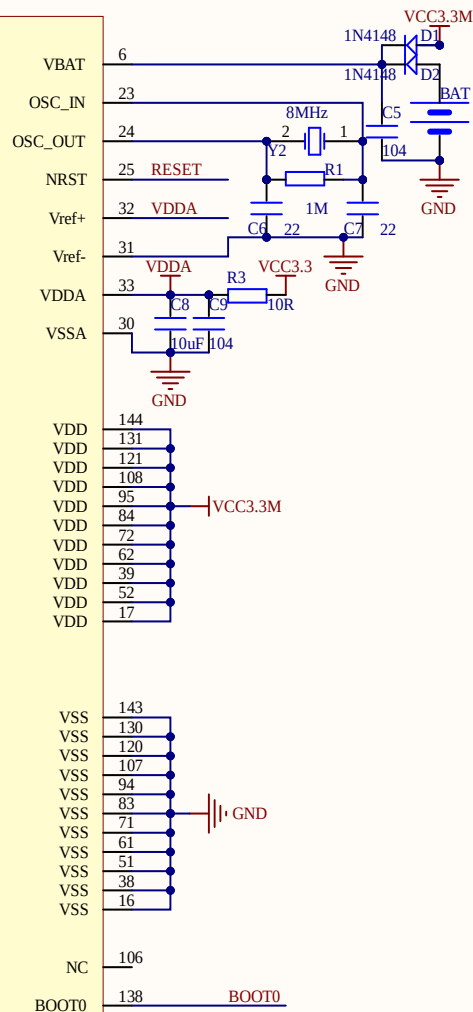


正点原子

## MCU\_B

| U1B          |      |     |                           |
|--------------|------|-----|---------------------------|
|              | PE0  | 141 | PE0/TIM4_ETR/FSMC_NBL0    |
|              | PE1  | 142 | PE1/FSMC_NBL1             |
|              | PE2  | 1   | PE2/TRACECK/FSMC_A23      |
| KEY1         | PE3  | 2   | PE3/TRACED0/FSMC_A19      |
| KEY0         | PE4  | 3   | PE4/TRACED1/FSMC_A20      |
| LED1         | PE5  | 4   | PE5/TRACED2/FSMC_A21      |
|              | PE6  | 5   | PE6/TRACED3/FSMC_A22      |
| FSMC_D4      | PE7  | 58  | PE7/FSMC_D4               |
| FSMC_D5      | PE8  | 59  | PE8/FSMC_D5               |
| FSMC_D6      | PE9  | 60  | PE9/FSMC_D6               |
| FSMC_D7      | PE10 | 63  | PE10/FSMC_D7              |
| FSMC_D8      | PE11 | 64  | PE11/FSMC_D8              |
| FSMC_D9      | PE12 | 65  | PE12/FSMC_D9              |
| FSMC_D10     | PE13 | 66  | PE13/FSMC_D10             |
| FSMC_D11     | PE14 | 67  | PE14/FSMC_D11             |
| FSMC_D12     | PE15 | 68  | PE15/FSMC_D12             |
|              |      |     |                           |
|              | PF0  | 10  | PF0/FSMC_A0               |
|              | PF1  | 11  | PF1/FSMC_A1               |
|              | PF2  | 12  | PF2/FSMC_A2               |
|              | PF3  | 13  | PF3/FSMC_A3               |
|              | PF4  | 14  | PF4/FSMC_A4               |
|              | PF5  | 15  | PF5/FSMC_A5               |
|              | PF6  | 18  | PF6/ADC3_IN4/FSMC_NIORD   |
|              | PF7  | 19  | PF7/ADC3_IN5/FSMC_NREG    |
| LIGHT_SENSOR | PF8  | 20  | PF8/ADC3_IN6/FSMC_NIOWR   |
| T_MOSI       | PF9  | 21  | PF9/ADC3_IN7/FSMC_CD      |
| T_PEN        | PF10 | 22  | PF10/ADC3_IN8/FSMC_INTR   |
| T_CS         | PF11 | 49  | PF11/FSMC_NIOS16          |
|              | PF12 | 50  | PF12/FSMC_A6              |
|              | PF13 | 53  | PF13/FSMC_A7              |
|              | PF14 | 54  | PF14/FSMC_A8              |
|              | PF15 | 55  | PF15/FSMC_A9              |
|              |      |     |                           |
| FSMC_A10     | PG0  | 56  | PG0/FSMC_A10              |
|              | PG1  | 57  | PG1/FSMC_A11              |
|              | PG2  | 87  | PG2/FSMC_A12              |
|              | PG3  | 88  | PG3/FSMC_A13              |
|              | PG4  | 89  | PG4/FSMC_A14              |
|              | PG5  | 90  | PG5/FSMC_A15              |
| NRF_IRQ      | PG6  | 91  | PG6/FSMC_INT2             |
| NRF_CS       | PG7  | 92  | PG7/FSMC_INT3             |
| NRF_CE       | PG8  | 93  | PG8                       |
|              | PG9  | 124 | PG9/FSMC_NE2/FSMC_NCE3    |
|              | PG10 | 125 | PG10/FSMC_NCE4_1/FSMC_NE3 |
| 1WIRE_DQ     | PG11 | 126 | PG11/FSMC_NCE4_2          |
| FSMC_NE4     | PG12 | 127 | PG12/FSMC_NE4             |
| OV_SDA       | PG13 | 128 | PG13/FSMC_A24             |
| OV_RRST      | PG14 | 129 | PG14/FSMC_A25             |
| OV_OE        | PG15 | 132 | PG15                      |

STM32F103ZET6\_AB

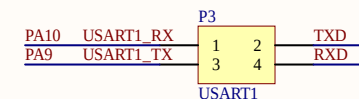


R7/R8 是为了维修方便而增加的2个电阻,大家实际设计电路板时候,可以不加这两个电阻。VCC3.3M 直接连接3.3V即可。

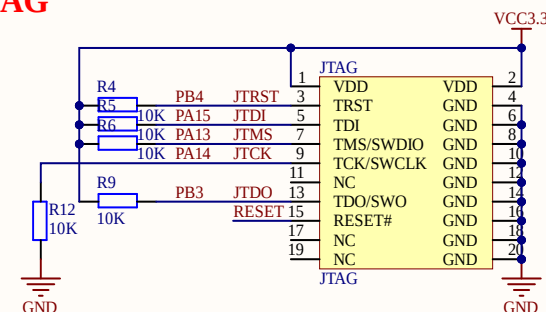
## IO

| P1       |    |    | P2       |    |    |
|----------|----|----|----------|----|----|
| PE0      | 1  | 2  | PB9      | 1  | 2  |
| PE2      | 3  | 4  | PE1      | 3  | 4  |
| PE4      | 5  | 6  | PE3      | 5  | 6  |
| PE6      | 7  | 8  | PE5      | 7  | 8  |
| PF0      | 9  | 10 | PC13     | 9  | 10 |
| PF2      | 11 | 12 | PF1      | 11 | 12 |
| PF4      | 13 | 14 | PF3      | 13 | 14 |
| PF6      | 15 | 16 | PF5      | 15 | 16 |
| PF8      | 17 | 18 | PF7      | 17 | 18 |
| PF10     | 19 | 20 | PF9      | 19 | 20 |
| PC1      | 21 | 22 | PC0      | 21 | 22 |
| PC3      | 23 | 24 | PC2      | 23 | 24 |
| PA1      | 25 | 26 | PA0      | 25 | 26 |
| PA5      | 27 | 28 | PA4      | 27 | 28 |
| PA7      | 29 | 30 | PA6      | 29 | 30 |
| PC5      | 31 | 32 | PC4      | 31 | 32 |
| PB1      | 33 | 34 | PB0      | 33 | 34 |
| PF11     | 35 | 36 | PB2      | 35 | 36 |
| PF13     | 37 | 38 | PF12     | 37 | 38 |
| PF15     | 39 | 40 | PF14     | 39 | 40 |
| PG1      | 41 | 42 | PG0      | 41 | 42 |
| PE8      | 43 | 44 | PE7      | 43 | 44 |
| PE10     | 45 | 46 | PE9      | 45 | 46 |
| PE12     | 47 | 48 | PE11     | 47 | 48 |
| PE14     | 49 | 50 | PE13     | 49 | 50 |
| PB10     | 51 | 52 | PE15     | 51 | 52 |
| GND      | 53 | 54 | PB11     | 53 | 54 |
| HEAD2*27 |    |    | HEAD2*27 |    |    |

## USB\_UART/USART1



## JTAG



Title:

ALIENTEK Elite STM32F103 Board

Author:

ATOM

Date:

2025/11/3

Revision:

\*

Size:

A4

File:

CORE\_EFG.SchDoc

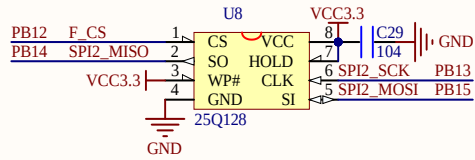
Version:

V2.6

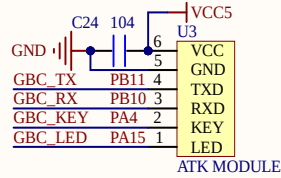


正点原子

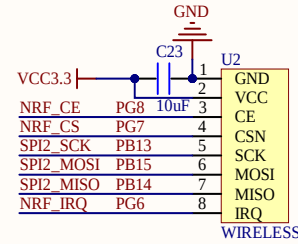
## SPI FLASH



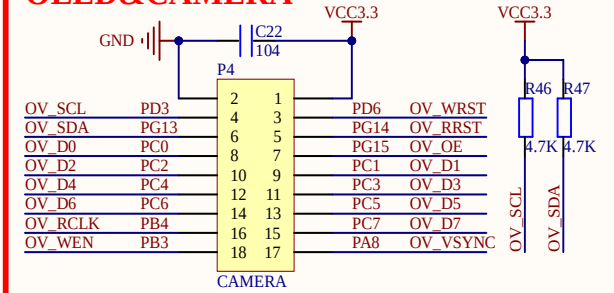
## ATK MODULE



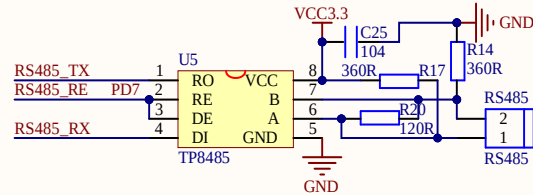
## WIRELESS



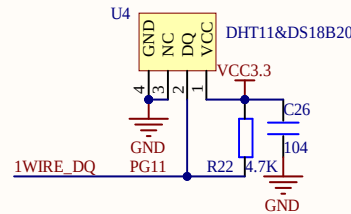
## OLED&CAMERA



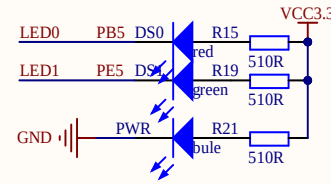
## RS485



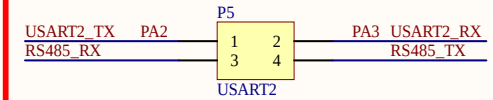
## TEMP&HUMI SENSOR



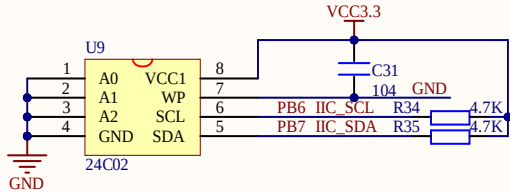
## LED



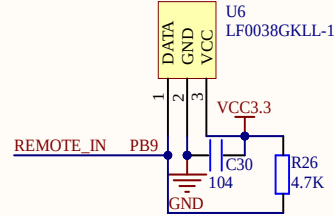
## RS485 SELECT



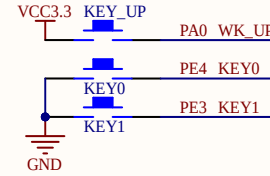
## EEPROM



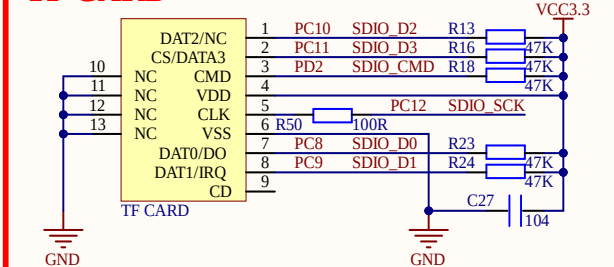
## REMOTE



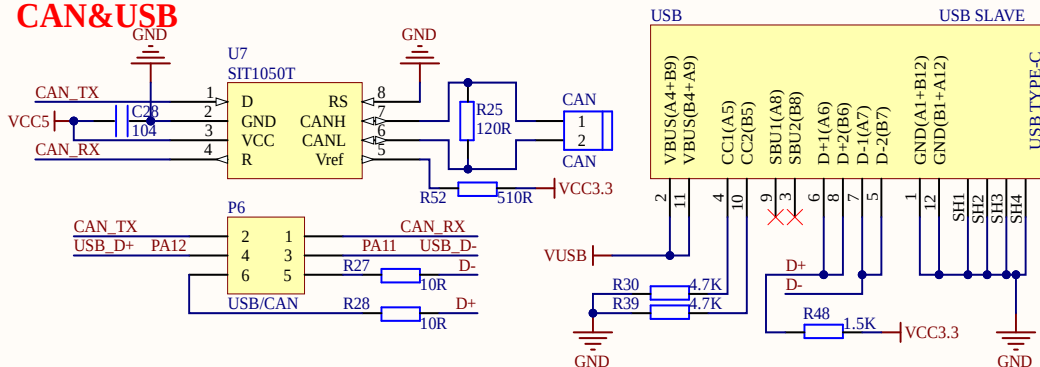
## KEY



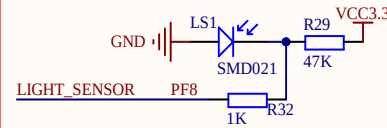
## TF CARD



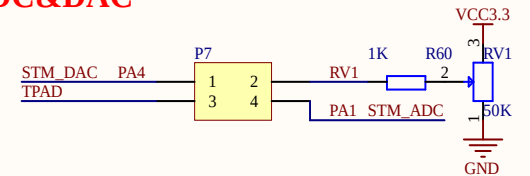
## CAN&USB



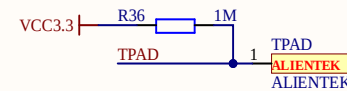
## LIGHT\_SENSOR



## ADC&DAC



## TOUCH\_KEY

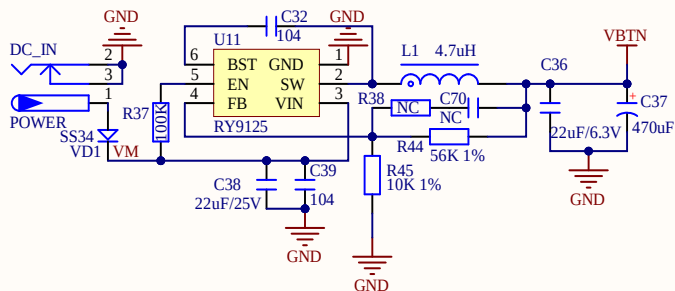


|           |                                |          |                |
|-----------|--------------------------------|----------|----------------|
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| Author:   | ATOM                           | File:    | DEVICES.SchDoc |
| Date:     | 2025/11/3                      | Version: | V2.6           |
| Revision: | 0                              |          |                |

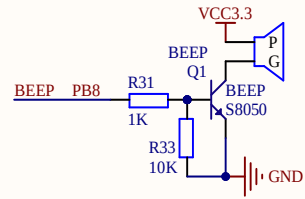


正点原子

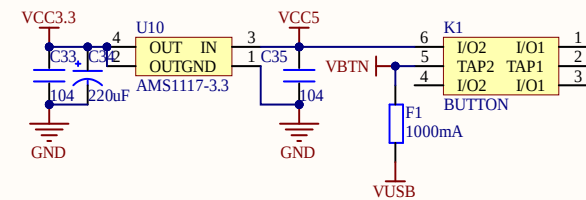
## DC POWER IN



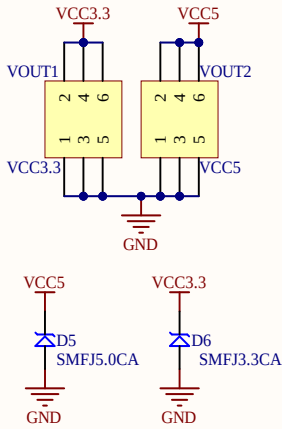
## BEEP



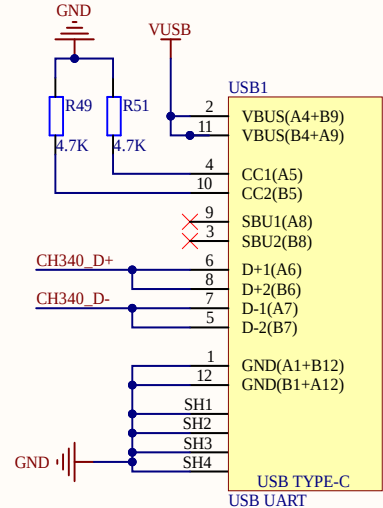
## 3.3V & POWER SWITCH



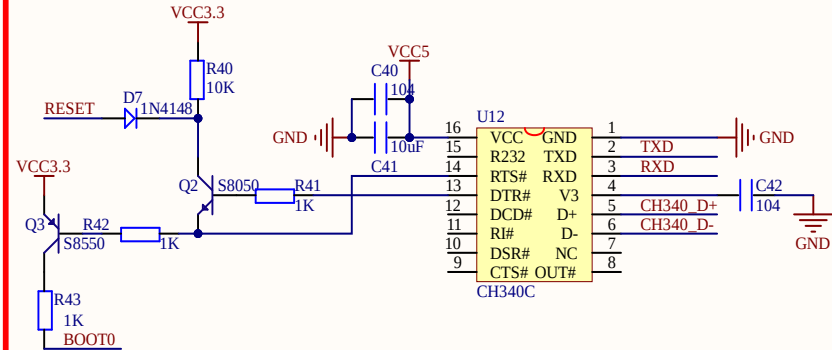
## ON BOARD POWER SOURCE



## USB USART



## USB POWER



|                                          |                                 |                                                                                               |
|------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------|
| Title:<br>ALIENTEK Elite STM32F103 Board |                                 | <br>正点原子 |
| Author:<br>ATOM                          | Size:<br>A4                     |                                                                                               |
| Date:<br>2025/11/3                       | File:<br>POWER&USB_USART.SchDoc |                                                                                               |
| Revision:<br>0                           | Version:<br>V2.6                |                                                                                               |

